FACULTY OF COMPUTING

SECD2613 SYSTEM ANALYSIS AND DESIGN
SECTION 02

SEMESTER 2 2025/2026

TOPIC:

PROJECT 1-2

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1.0 Introduction

Autism, or Autism Spectrum Disorder (ASD), refers to a group of lifelong neurodevelopmental conditions (Chan, 2025). In recent years, awareness of autism has increased significantly, leading to a greater recognition of the importance of early detection and continuous support for individuals with autism. Early diagnosis of autism is crucial for timely intervention and improved long-term outcomes (Okoye et al., 2023). This review aims to explore some of its signs and symptoms, look into some diagnostic tools, and analyze the benefits and risks associated with an early diagnosis of autism (Okoye et al., 2023). However, despite early identification being important, many individuals still experience difficulties in everyday functioning because autism can affect how they manage routines, adapt to changes, and maintain consistent daily schedules, highlighting the need for structured support systems to help them navigate daily life more effectively.

This project focuses on the development of an Autism Early Screening Support System and Autism Daily Routine Management System. The system is designed to assist parents and healthcare providers in identifying early signs of autism through a user-friendly screening process while also helping individuals with autism to manage their daily activities effectively. By integrating both features in one single platform, the system aims to provide a more comprehensive support solution for autism care and assistance.

The Autism Early Screening Support System is intended to simplify the preliminary screening process by providing questionnaires and guidance based on behavioral indicators associated with ASD. Although the system does not replace professional medical diagnosis, it serves as an early support tool that encourages timely consultation with specialists. In fact, early screening can reduce delays in diagnosis and increase the opportunities for early intervention programs that contribute positively to a child's development.

Daily routine management is important for individuals with Autism Spectrum Disorder (ASD) because it provides structure, consistency, and predictability in everyday life. Many autistic individuals prefer familiar and repetitive activities, as structured routines can help reduce anxiety, stress, and behavioral difficulties while improving emotional regulation and independence. Daily routines commonly include scheduled activities such as meals, learning sessions, therapy, physical activities, and sleep patterns. Visual schedules, reminders, and organized task sequencing are often used to help autistic individuals manage transitions between activities more effectively. According to Russell et al. (2025), delayed autism diagnosis may limit access to early intervention and support systems that are important for developing effective coping strategies and daily management skills [7]. Therefore, establishing consistent daily routines can significantly improve communication, adaptive functioning, and overall quality of life for individuals with autism.

The integration of technology into the autism support system demonstrates the growing role of digital solutions in healthcare. Through this project, the proposed system aims to provide an accessible, practical, and supportive platform that enhances awareness, promotes early intervention, and improves daily living management for individuals with autism and their parents or caregivers.

2.0 Background Study

Early identification of autism is crucial as it enables timely intervention that can significantly improve developmental outcomes such as their communication, behaviour and social skills. According to Okoye et al. (2023), early diagnosis of autism allows parents and healthcare professionals to provide appropriate support at an earlier stage which can enhance long term progress [2]. However, many cases are still identified late due to limited awareness of early warning signs and the lack of accessible screening support tools for caregivers.

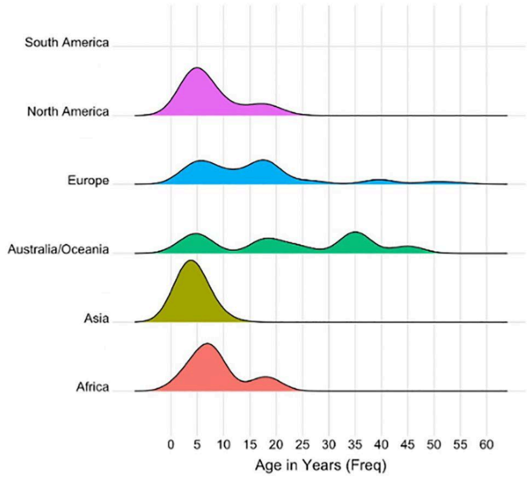


Figure 1. Ridge plot of autism late diagnosis frequency based on world continents. Reprinted/adapted from Who, when, where, and why: A systematic review of "late diagnosis" in autism, by A. S. Russell, T. C. McFayden, M. McAllister, et al., 2025, Autism Research, 18(1), 22–36. https://doi.org/10.1002/aur.3278

The ridge plot illustrates the distribution of autism diagnosis ages across different world continents, showing that most diagnoses occur during childhood, particularly between 4 and 10 years of age. Africa, Asia, and North America display strong peaks in early childhood, indicating that autism is commonly identified during school-age years, although smaller secondary peaks in adolescence suggest the presence of delayed diagnoses. Europe and Australia/Oceania exhibit broader distributions extending into adulthood, reflecting higher rates of late diagnosis and suggesting better access to adult diagnostic services, increased autism awareness, and more developed healthcare systems. In contrast, the absence of a visible distribution for South America may indicate insufficient or limited data. Overall, the plot highlights global differences in autism screening with developed regions showing greater recognition of autism across all age groups.

Autism daily routine management plays an important role in supporting individuals with Autism Spectrum Disorder (ASD) by promoting structure, predictability, and emotional stability in everyday life. Consistent routines help reduce anxiety and behavioral challenges because many autistic individuals prefer familiar and repetitive activities. Daily routine management may include scheduled sleeping patterns, meal times, and hygiene practices. Visual schedules, reminders, and structured task sequencing are commonly used strategies to improve independence and assist individuals in completing daily activities more effectively. Maintaining a predictable routine also supports communication development, emotional regulation, and adaptive functioning in both children and adults with autism.

According to Russell et al. (2025), delayed autism diagnosis may affect access to early interventions and routine-based support systems, which are essential for long-term developmental outcomes [7]. Individuals diagnosed later in life may experience greater difficulties in establishing effective coping strategies and structured daily management due to years of insufficient support needs. Therefore, implementing personalized daily routines after diagnosis can significantly improve quality of life, reduce stress, and enhance social and cognitive functioning. Families, caregivers, educators, and healthcare professionals all play a crucial role in developing structured environments that accommodate the individual needs and sensory preferences of autistic individuals.

Existing autism support solutions include screening tools, mobile applications, and behaviour tracking systems. However, many of these systems operate separately and focus either on early screening or daily routine management rather than integrating both functions into a single platform. Furthermore, some of the existing systems lack autism-friendly features which are important for improving usability and effectiveness. Therefore, there is a need for an integrated system that combines early autism screening with daily routine management features.

3.0 Problem Statement

Problem 1 - Late or missed early diagnosis
Parents have no structured, self-service platform to identify early signs of ASD, leaving them dependent on informal observation or lengthy referral processes. This directly delays diagnosis and causes many children to miss the critical early intervention window that is essential for better long-term development outcomes.

Problem 2 - Absence of structured daily routine support
Individuals with ASD consistently struggle with feeding, sleep and hygiene routines, yet no desiccated digital solution exists to address this. General scheduling applications fail to accommodate sensory sensitivities, visual communication needs, or the gradual behavioural progression required for individuals with ASD, making them ineffective for this specific user group.

Problem 3 - Lack of an Integrated Platform for Parents and Healthcare Providers
Screening and daily routine management are handled through separate, disconnected channels with no single platform bridging both. This fragmentation is further compounded by healthcare providers having to rely on caregiver reports during consultation rather than structured tracked data, reducing the quality of clinical decisions and intervention planning.

4.0 Proposed Solution

To address the identified problems, this project proposed the development of a unified web-based platform consisting of two integrated modules. The first is the Autism Early Screening Support System, which provides parents with a structured, guided questionnaire based on validated ASD behavioural indicators to support early identification and timely referral to specialists. Second, the Autism Daily Routine Management System, which offers visual, sensory-friendly schedules across feeding, sleep, and hygiene routines to help individuals with ASD build consistency and independence in their daily activities. By combining these two modules within a single platform accessible to both parents and healthcare providers, the system eliminates the fragmentation of the current support channel and ensures that clinicians have access to consolidated, structured data to support better-informed consultation and intervention planning.

4.1 Technical Feasibility

The system will be hosted on a standard web server that can be accessed through any browser on a computer or mobile device without requiring installation. Although the development team is still in the process of learning some of these technologies, sufficient resources such as online documentation, tutorials, and lecturer guidance are available to support the team throughout the development process. At this stage, the project has progressed through the planning, analysis, and design phases of the System Development Life Cycle (SDLC). The planning phase involved identifying the system objectives, scope, and feasibility. The analysis phase covered requirements gathering, problem identification, and current system evaluation. The design phase focused on producing system diagrams including the Data Flow Diagram, process flows, and interface wireframes. The system has not yet proceeded to the implementation, testing, or maintenance phases, as development is currently ongoing within the design stage. .

4.2 Operational Feasibility

The proposed system is designed to be simple and easy to use for both parents and healthcare providers. Parents can access the screening questionnaire and manage their child's daily routines through a clean and straightforward interface that does not require any technical knowledge or training. Healthcare providers, on the other hand, can view structured screening results and routine records submitted by parents, which helps them make better decisions during consultations without changing their existing workflow. Since the system is browser-based, users can access it from any device at any time, making it convenient for everyday use. With its simple design and clear benefits for both user groups, the system is expected to be well accepted and easy to operate.

4.3 Economical Feasibility

Estimated Costs:		**Expected Benefits:**	
Web Developer	RM 15,000	Savings	RM 20,000
Software	RM 10,000	Increase productivity	RM 30,000
Hardware	RM 3,000		
Training	RM 5,000		
Consultant	RM 2,000		
Upgrade	RM 3,000		
Maintenance	RM 2,000 per month		
IS Support	RM 700 per month		

Assumptions:

Discount rate	10%
Sensitivity factor(cost):	0.9
Sensitivity factor(benefit):	1.2
Annual increment(cost):	5%
Annual increment(benefit):	10%

4.3.1 CBA

CRITERIA	YEAR 0	YEAR 1	YEAR 2	YEAR 3
DEVELOPMENT COST				
Web Developer	13,500	-	-	-
Software	9,000	-	-	-
Hardware	2,700	-	-	-
Training	4,500	-	-	-
Consultant	1,800	-	-	-
TOTAL	31,500	-	-	-
OPERATION COST				
Upgrade	-	2,700	2,835	2,976.75
Maintenance	-	21,600	22,680	23,814
IS Support	-	7,560	7,938	8,334.90
TOTAL	-	31,860	33,453	35,125.65
Present Value (PV) PV = Payment x (1/(1+C)^n)	-	31,860 x (1/(1+0.1)^1) =28,963.636	33,453 x (1/(1+0.1)^2) =27,647.107	35,125.65 x (1/(1+0.1)^3) =26,390.421
Accumulated Cost	-	60,463.636	88,110.743	114,501.164
BENEFITS				
Savings	-	24,000	26,400	29,040
Increase productivity	-	36,000	39,600	43,560
TOTAL	-	60,000	66,000	72,600
Present Value (PV) PV = Payment x (1/(1+C)^n)	-	60,000 x (1/(1+0.1)^1) =54,545.455	66,000 x (1/(1+0.1)^2) =54,545.455	72,600 x (1/(1+0.1)^3) =54,545.455
Accumulated	-	54,545.455	109,090.91	163,636.365

Benefits				
Gain or Loss	-	(5,918.181)	20,980.167	49,135.201
Profitable Index (PI)	1.56			

Economic feasibility is assessed through a Cost-Benefit Analysis (CBA) projected over three years using a discount rate of 10%, a cost sensitivity factor of 0.9, a benefit sensitivity factor of 1.2, a 5% annual cost increment, and a 10% annual benefit increment. The total one-time development cost at Year 0 is RM 31,500, covering web developer fees, software, hardware, training, and consultant charges. Annual recurring costs begin at RM 31,860 in Year 1, covering maintenance, IS support, and system upgrades. The expected annual benefit starts at RM 60,000 in Year 1, derived from operational savings and increased productivity gains, and grows progressively over the three-year period. Based on the CBA calculation, the Profitability Index (PI) is **1.56**, indicating that for every RM 1.00 invested, the system returns RM 1.56 in present value benefits. Since PI is greater than 1.0, the proposed system is concluded to be economically feasible and is recommended to proceed to the next phase of development.

5.0 Objectives

- To identify problems and opportunities related to autism early screening and daily routine management among parents, autistic children and NGOs.
- To conduct data collection and analyze user requirements on their goals, data and procedure involved through interview, questionnaires and observation.
- To analyze the functional and non-functional requirements of the proposed Autism Early Screening Support System.
- To design a digitalized Autism Early Screening Support System that improves accessibility, early screening processes and supports autism daily routine management.
- To prepare system analysis and design documentation such as data flow diagrams (DFD) and system proposal for the proposed system.

6.0 Scope of the Project

This project focuses on the planning, analysis and design of an Autism Early Screening Support System for parents, autistic children and a non-profit organization (NGO). The proposed system is intended to improve the current autism screening process and add autism daily routine management by providing a more organized and centralized platform. The scope of the project covers the following areas.

Included scope:

- Conducting background research related to autism early screening and current support practices.
- Identifying existing problems, opportunities and user requirements through system analysis activities.
- Gathering and analyzing information from users using interviews, questionnaires and observations.
- Preparing feasibility analysis including technical, operational and economic.
- Preparing project planning documentation including Work Breakdown Structure (WBS), PERT Diagram and Gantt Chart.
- Managing project development activities using Github for task tracking.
- Producing system analysis documentation such as Flowchart, Data Flow Diagram (DFD) and process models for current systems.
- Producing system design documentation including DFD, CRUD Matrix, Event Response Table, Structure Chart and System Architecture.
- Creating system wireframes by developing non-working prototypes include input and output design.

Not included scope:

- Full system development and coding implementation
- System testing, maintenance and evaluation
- System implementation and deployment

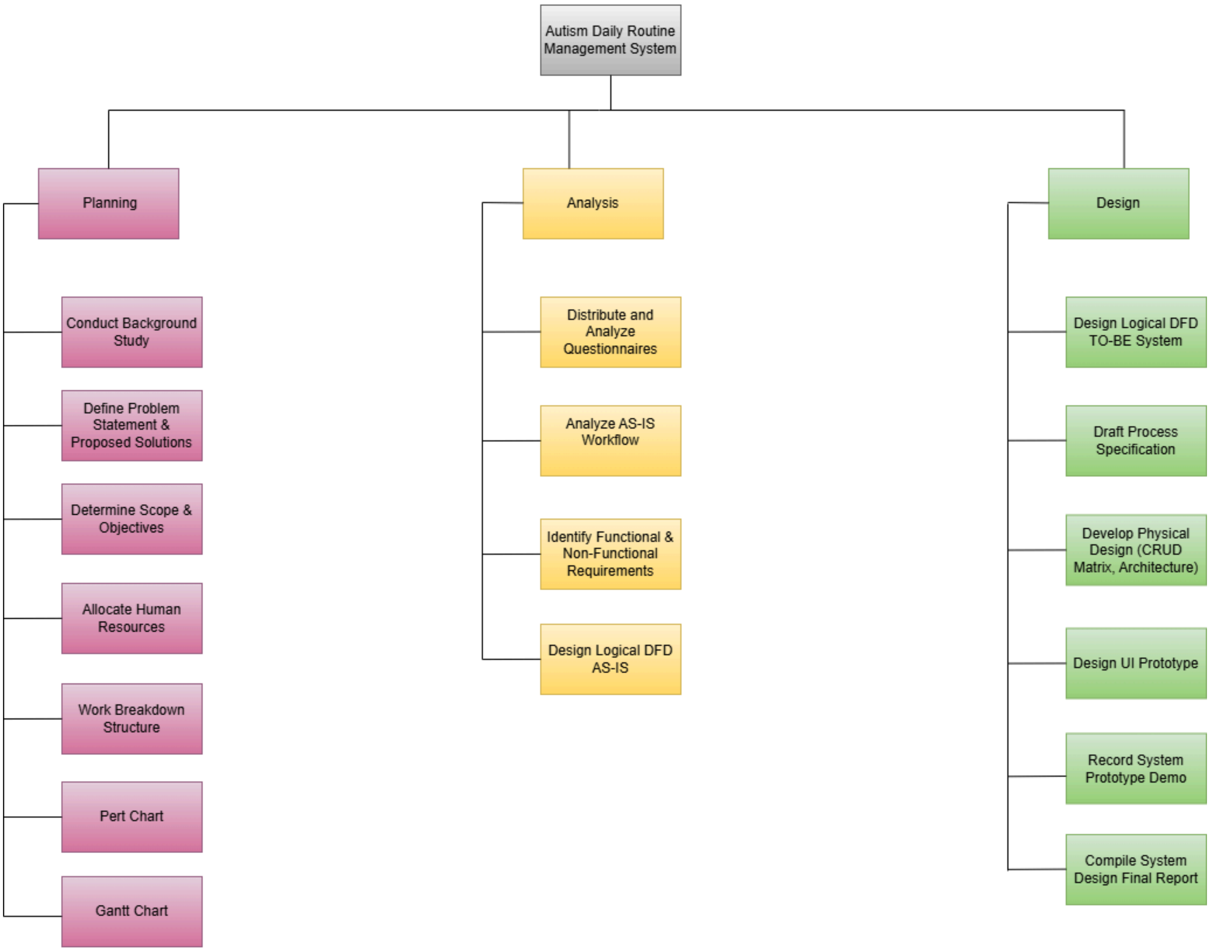
7.0 Project Planning

7.1 Human Resources

Person in charge	Responsibility	Sub-Module
AKMAL RAFIQUE BIN AHMAD RAPHAIE (Project Manager)	**Planning Branch:** • Allocate Human Resources • Work Breakdown Structure • PERT chart • Gantt chart **Analysis Branch:** • Distribute and Analyze Questionnaires **Design Branch:** • Design registration system • Compile System Design Final Report (Project Closure/Sign-off)	Authentication & Registration Module
FARAH ADILAH BINTI AZMAN (UI/UX Designer)	**Planning Branch:** • Conduct Background Study • Gantt chart **Analysis Branch:** •Analyze AS-IS Workflow (Routine processes) **Design Branch:** • Design routine management UI prototype • Record System Prototype Demo	Daily Routine Management Module
NURAIN NABILAH BINTI AZMAN (Systems Analyst)	**Planning Branch:** • Define Problem Statement & Proposed Solutions • Gantt chart **Analysis Branch:** • Design Logical DFD AS-IS **Design Branch:** • Design Logical DFD TO-BE System • Develop Physical Design (CRUD Matrix, Architecture)	Information Sharing & Database Module
MURSYIDAH BINTI JAHIDI	**Planning Branch:** • Determine Scope & Objectives • Gantt chart **Analysis Branch:** • Identify Functional & Non-Functional	Early Screening Assessment Module

(Requirements Engineer / QA)	Requirements **Design Branch:** • Draft Process Specification for screening assessment	

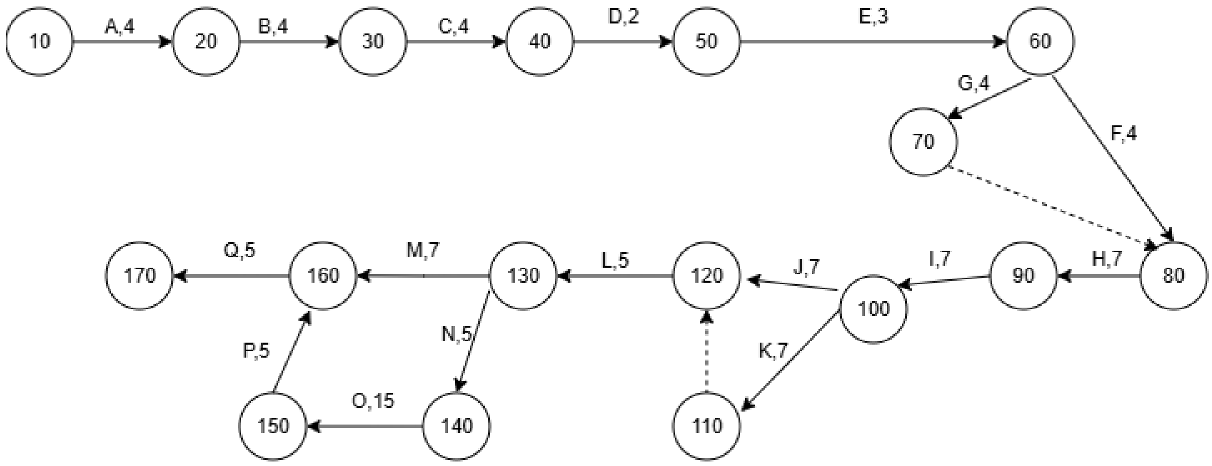
7.2 Work Breakdown Structure (WBS)



7.3 PERT Chart

Activity		Predecessor	Duration (days)
Planning phase			
A	Conduct Background Study	NONE	4
B	Define Problem Statement & Proposed Solutions	A	4
C	Determine Scope & Objectives	B	4
D	Allocate Human Resources	C	2
E	Work Breakdown Structure	D	3
F	Pert Chart	E	4
G	Gantt Chart	E	4
Analysis phase			
H	Distribute and Analyze Questionnaires	F,G	7
I	Analyze AS-IS Workflow	H	7
J	Identify Functional & Non-Functional Requirements	I	7
K	Design Logical DFD AS-IS	I	7
Design phase			
L	Design Logical DFD TO-BE System	J,K	5
M	Draft Process Specification	L	7
N	Develop Physical Design (CRUD Matrix, Architecture)	L	5
O	Design UI Prototype	N	15
P	Record System Prototype Demo	O	5
Q	Compile System Design Final Report	M,P	5

Pert Chart Reference Table



Pert Chart

All Paths
Path 1: A ➔ B ➔ C ➔ D ➔ E ➔ F ➔ H ➔ I ➔ J ➔ L ➔ M ➔ Q
Length: 4 + 4 + 4 + 2 + 3 + 4 + 7 + 7 + 7 + 5 + 7 + 5 = 59 days

Path 2: A ➔ B ➔ C ➔ D ➔ E ➔ G ➔ H ➔ I ➔ J ➔ L ➔ M ➔ Q
Length: 4 + 4 + 4 + 2 + 3 + 4 + 7 + 7 + 7 + 5 + 7 + 5 = 59 days

Path 3: A ➔ B ➔ C ➔ D ➔ E ➔ F ➔ H ➔ I ➔ K ➔ L ➔ M ➔ Q
Length: 4 + 4 + 4 + 2 + 3 + 4 + 7 + 7 + 7 + 5 + 7 + 5 = 59 days

Path 4: A ➔ B ➔ C ➔ D ➔ E ➔ G ➔ H ➔ I ➔ K ➔ L ➔ M ➔ Q
Length: 4 + 4 + 4 + 2 + 3 + 4 + 7 + 7 + 7 + 5 + 7 + 5 = 59 days

Path 5: A ➔ B ➔ C ➔ D ➔ E ➔ F ➔ H ➔ I ➔ J ➔ L ➔ N ➔ O ➔ P ➔ Q
Length: 4 + 4 + 4 + 2 + 3 + 4 + 7 + 7 + 7 + 5 + 5 + 15 + 5 + 5 = 77 days

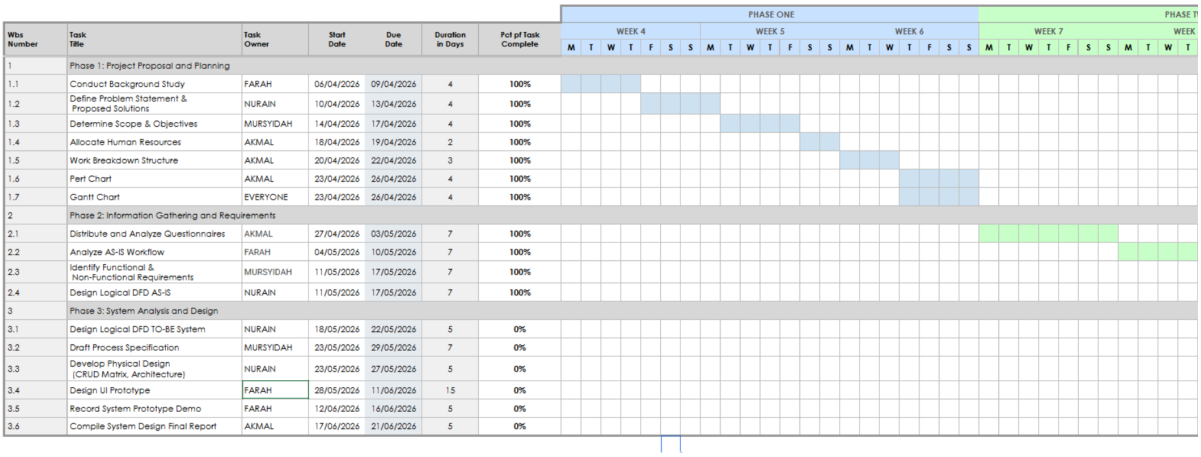
Path 6: A ➔ B ➔ C ➔ D ➔ E ➔ G ➔ H ➔ I ➔ J ➔ L ➔ N ➔ O ➔ P ➔ Q
Length: 4 + 4 + 4 + 2 + 3 + 4 + 7 + 7 + 7 + 5 + 5 + 15 + 5 + 5 = 77 days

Path 7: A ➔ B ➔ C ➔ D ➔ E ➔ F ➔ H ➔ I ➔ K ➔ L ➔ N ➔ O ➔ P ➔ Q
Length: 4 + 4 + 4 + 2 + 3 + 4 + 7 + 7 + 7 + 5 + 5 + 15 + 5 + 5 = 77 days

Path 8: A ➔ B ➔ C ➔ D ➔ E ➔ G ➔ H ➔ I ➔ K ➔ L ➔ N ➔ O ➔ P ➔ Q
Length: 4 + 4 + 4 + 2 + 3 + 4 + 7 + 7 + 7 + 5 + 5 + 15 + 5 + 5 = 77 days

The critical path is always the path with the longest total duration, as it determines the minimum amount of time required to complete the entire project.
Since the critical path is the longest path through the network diagram, Path 5,6,7,8 is the critical path for this Project.

7.4 Gantt Chart



Wbs Number	Task Title	Task Owner	Start Date	Due Date	Duration in Days	Pct pf Task Complete
1	Phase 1: Project Proposal and Planning					
1.1	Conduct Background Study	FARAH	06/04/2026	09/04/2026	4	100%
1.2	Define Problem Statement & Proposed Solutions	NURAIN	10/04/2026	13/04/2026	4	100%
1.3	Determine Scope & Objectives	MURSYIDAH	14/04/2026	17/04/2026	4	100%
1.4	Allocate Human Resources	AKMAL	18/04/2026	19/04/2026	2	100%
1.5	Work Breakdown Structure	AKMAL	20/04/2026	22/04/2026	3	100%
1.6	Pert Chart	AKMAL	23/04/2026	26/04/2026	4	100%
1.7	Gantt Chart	EVERYONE	23/04/2026	26/04/2026	4	100%
2	Phase 2: Information Gathering and Requirements					
2.1	Distribute and Analyze Questionnaires	AKMAL	27/04/2026	03/05/2026	7	100%
2.2	Analyze AS-IS Workflow	FARAH	04/05/2026	10/05/2026	7	100%
2.3	Identify Functional & Non-Functional Requirements	MURSYIDAH	11/05/2026	17/05/2026	7	100%
2.4	Design Logical DFD AS-IS	NURAIN	11/05/2026	17/05/2026	7	100%
3	Phase 3: System Analysis and Design					
3.1	Design Logical DFD TO-BE System	NURAIN	18/05/2026	22/05/2026	5	0%
3.2	Draft Process Specification	MURSYIDAH	23/05/2026	29/05/2026	7	0%
3.3	Develop Physical Design (CRUD Matrix, Architecture)	NURAIN	23/05/2026	27/05/2026	5	0%
3.4	Design UI Prototype	FARAH	28/05/2026	11/06/2026	15	0%
3.5	Record System Prototype Demo	FARAH	12/06/2026	16/06/2026	5	0%
3.6	Compile System Design Final Report	AKMAL	17/06/2026	21/06/2026	5	0%

Gantt Chart

Reference Link:
https://docs.google.com/spreadsheets/d/1eNbYAx_T9S3i_cAb9WYNlXsJpQKBlSN2hB0NA6fTe_k/edit?usp=sharing

8.0 Benefit and Overall Summary of Proposed System

The proposed autism early screening support system provides several benefits to its users, especially parents, autistic children and the NGO as the system owner. First, it improves early awareness by allowing users to conduct preliminary autism screening through the M-CHAT-R tool without the need to visit a clinic or hospital for the screening process. The screening results will be generated immediately after the questionnaire is submitted. This can help parents gain quick insight about their child developmental status. Second, the system increases accessibility and convenience as it is available on mobile application and web-based platforms which enable users, especially parents, to perform the screening anytime and anywhere. Third, it supports better child routine management through the daily routine module, helping parents organize structured daily activities such as meals, sleep and hygiene routines. In addition, the notification and reminder feature ensures that users do not miss follow-up activities or new information about autism. Lastly, the reporting and analytics feature helps parents and NGO monitor screening summaries and view autism related statistics. This can help increase awareness and understanding of autism through simple data visualization and reports.

Overall, the proposed system is developed to address several major issues related to autism early screening and daily routine management. Currently, many parents experience delays in identifying early signs of Autism Spectrum Disorder (ASD) due to the lack of a structured and accessible screening platform. In addition, general scheduling apps do not fully support the daily routine needs of autistic children, also screening and routine management are often handled separately without a centralized system. To overcome these problems, the proposed system integrates autism early screening, reporting and analytics, notification and remainder feature, and daily routine management into a single system. The system allows users to conduct preliminary screening and receive instant results, while also helping them manage their autistic child daily routine more effectively. However, the system does not replace professional medical diagnosis or treatment. It serves as a supportive system that helps improve early detection and awareness, in which users are still required to seek proper medical consultation for confirmed diagnosis and get further treatment.

9.0 Information Gathering Process

The main objective of the information gathering phase was to clearly understand the current AS-IS workflow of how parents, caregivers and special education teachers handle daily routines especially feeding, sleeping and hygiene of children with Autism Spectrum Disorder (ASD), as well as their experiences with early screening. By reviewing this manual process, the project team was able to identify key problems, systemic inefficiencies and important user needs. This data served as the basis for establishing the functional and non-functional requirements of the proposed digital TO-BE system.

9.1 Method Used

To gather detailed, accurate, and quantitative data, the team chose to distribute an online questionnaire via Google Forms as the primary data collection method. This quantitative approach was chosen for several reasons:

- Usually, Caregivers and special education teachers of children with ASD have busy and changing schedules. Online questionnaires allow them to respond whenever they are free without needing to attend a physical session.
- Using multiple-choice questions, linear scales, and targeted short answers will produces more clearer and easier data. This data will makes it easier for the team to analyze the responses and identify the actual needs of users for the proposed system.
- topics related to developmental delays and behavioral difficulties can be sensitive for some of the respondents. Anonymous online forms provide a more comfortable environment for respondents to share their experiences and challenges without being judged.

In addition to the quantitative questionnaire, the team also conducted an interview to gather qualitative insights. An interview was held with a subject matter expert, TS. Nurasyikin, during the data collection phase. This qualitative approach was chosen to complement the survey for several reasons:

- This interview allowed the team to explore the detailed processes of existing manual workflows and the main difficulties caregivers experience on a daily basis.
- The session provided the opportunity to ask additional follow up questions and better understand complicated issues, such as the exact reasons why current reporting formats fail and how the data is lost during transitions between caregivers.
- Engaging directly with a subject matter expert offered more strategic, high level perspectives on system requirements, ensuring the proposed digital solution matches real user needs, professional standards, and realistic problem solving strategies.

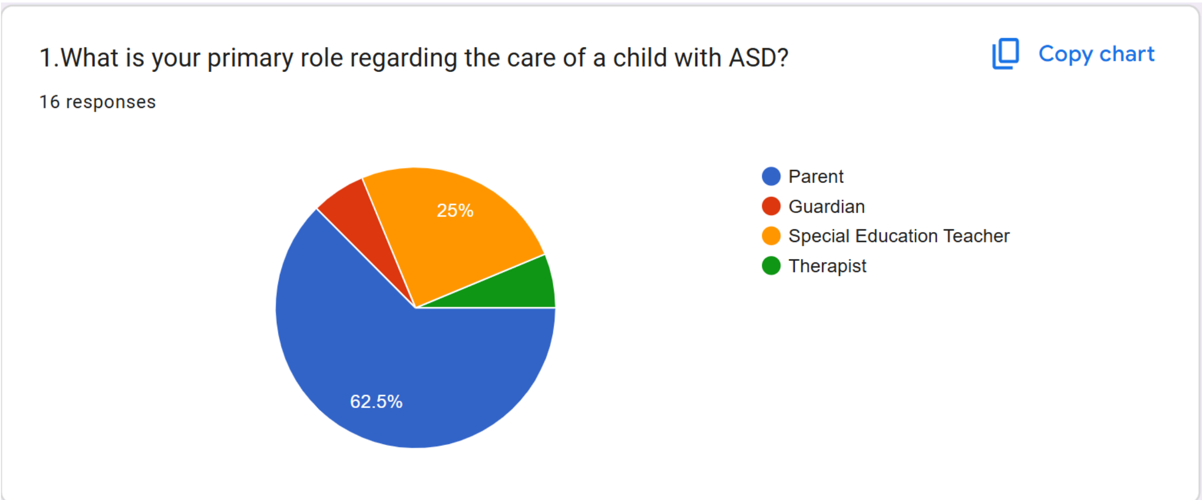
9.2 Summary of Method

Questionnaires

To thoroughly understand stakeholder needs, the project team developed and distributed an 18 question survey. A total of 16 valid responses were collected from the target group, which included parents, guardians, and special education professionals.

The questionnaire was carefully designed to examine both the current challenges in manual tracking systems (AS-IS) and the expected features of the proposed digital solution TO-BE. For clarity and focus in this documentation, only 7 key questions were selected for detailed analysis. These were chosen because their results had the most direct impact on the system design, particularly in defining core requirements such as an automated screening calculator, interactive visual timers, and a customizable, sensory friendly interface.

Overall, the feedback from these 16 respondents provides a strong, data-driven foundation for the system design and supports the development of the Data Flow Diagrams and architectural decisions discussed in the following sections.



To identify the main users involved in the system's system environment, the survey captured a total of 16 responses regarding the participants' primary roles in caring for a child with Autism Spectrum Disorder (ASD). The data reveals that parents are the most respondents of the target audience, accounting for 62.5% of the total responses. Special education teachers represent the second largest stakeholder group at 25%, while guardians and therapists form the remaining percentage, split evenly at 6.25% each. From a system analysis perspective, this distribution indicates that the "TO-BE" system's system requirements must focus mainly on the needs, daily processes, and ease of use of parents and educators, who together make up 87.5% of the core user base.

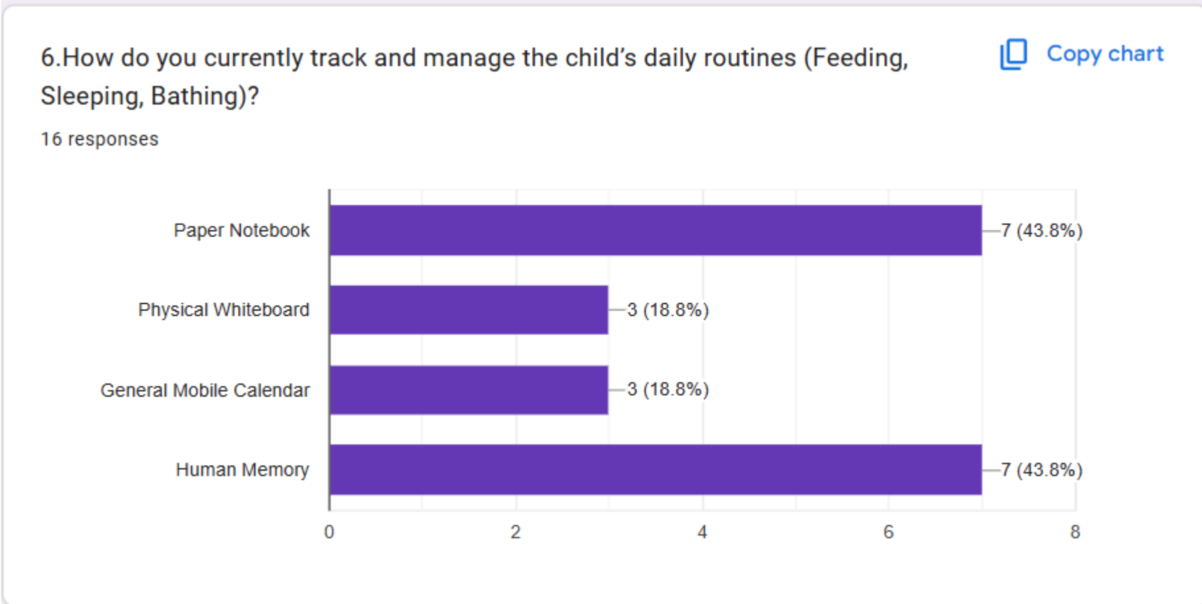
5.Briefly describe your current manual process if you wanted to check for early signs of autism.

16 responses

Summarize responses

- I mostly search on Google for symptoms and read parenting forums, but the information is overwhelming and confusing.
- I use a printed paper schedule on the wall, but I have to physically point to it multiple times. Sometimes I use my phone alarm.
- I manually record unusual behaviors or sensory sensitivities in a notebook for future reference.
- I wait months for a pediatrician appointment to ask them directly.
- look at their behavior and detect for any abnormal action. If critical i would go to clinic and check with specialist.
- search Google for symptoms and read parenting forums, but the information is overwhelming and confusing.
- check for their daily behavior and try to understand their need. Look for their emotions on doing something. whether they are happy, sad or mad at something.
- compare with other people growth, and if my child shows any abnormal behavior I will go to specialist and get my child diagnosed
- I ask teachers or family members if they notice similar developmental or communication issues.
- I usually search online for autism symptoms and compare them with the child's behavior.
- Going to the nearest Government Health Clinic for diagnosis
- I would start by observing a child's behavior and development over time. I'd look for signs such as limited eye contact, delayed speech, difficulty responding to their name.
- I just bring it up during normal checkups, but there is no formal test done at home.
- I ask other parents or my child's kindergarten teacher if they notice anything different

When asked to describe how they currently check for early signs of autism, respondents shared that the process is quite scattered and sometimes frustrating. Many parents said they usually start by searching online through Google or parenting forums to compare their child's behaviour with common autism symptoms, but they often find the information overwhelming and confusing because there is too much and it is not always clear. Others rely on daily observation, paying attention to things like eye contact, speech development, and how the child responds when their name is called, and some even jot down notes in notebooks to track unusual behaviour over time. A few also ask people around them, such as teachers, family members,or other parents, for their opinions. However, some respondents mentioned that they do not use any structured method at all and instead wait for clinic appointments or bring up their concerns during routine checkups. Overall, these responses show that current practices are unstructured and inconsistent, highlighting the need for a simpler and more organised digital screening tool.



To understand the current manual process, caregivers were asked how they manage and track daily routines such as feeding, sleeping, and bathing. The results show that most caregivers still depend on scattered and unorganized traditional methods rather than a proper digital system. About 43.8% use paper notebooks to record information, while 18.8% rely on a physical whiteboard. Another 18.8% use general mobile calendars together with memory, and the remaining 43.8% depend entirely on remembering mentally. These findings are important for the AS-IS analysis because they highlight how inconsistent and difficult to manage the current process is. Most caregivers are still handling important and important daily routines using manual tools or mental tracking, which can easily lead to forgotten information, errors, or difficulty sharing records with others. This clearly shows the need for a centralized system that can store and manage all information in one place more effectively and more effectively.

8.Briefly describe your current step-by-step manual process for getting the child to complete their most difficult routine.

16 responses

✦ Summarize responses

I write the routine on a whiteboard in the bathroom. I give verbal instructions, wait for them to finish, and wipe the board clean.

I use a printed paper schedule on the wall, but I have to physically point to it multiple times. Sometimes I use my phone alarm.

I break the routine into smaller tasks, praise every completed step, and avoid rushing the child.

break into smaller task and guide them to complete it.

I draw the steps (undress, soap, wash) on a whiteboard. We cross them out together with a red pen.

Observe how they complete the difficult task. Guide them for each task and help them if they can't complete it. be patient with them and treat them with love.

let them be free to do anything they want at first. then come out with a gentle remainder that they need to complete their routine and invite them to complete it together.

I use a printed picture schedule and point to each step while helping the child complete the routine.

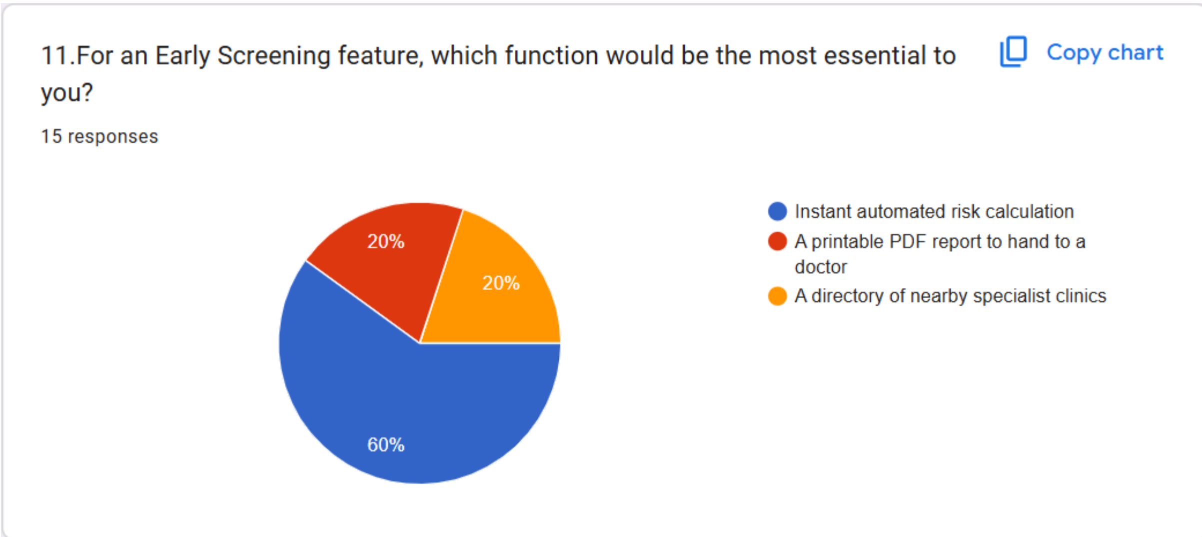
I prepare all the items needed beforehand, explain the task slowly, and reward the child with praise afterward.

I usually remind my child several times verbally, then guide them step-by-step until the routine is completed.

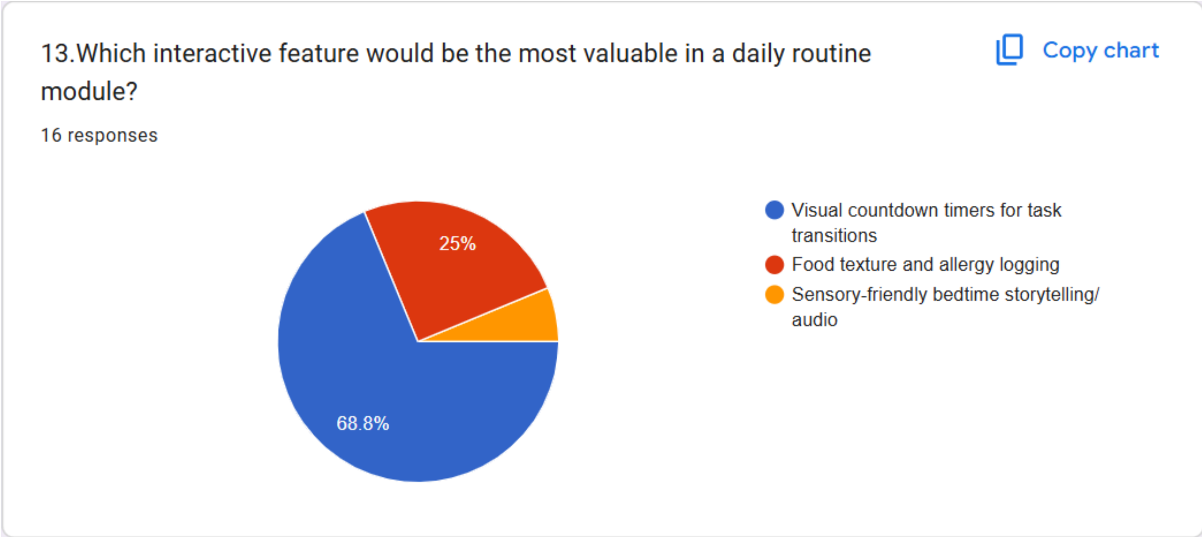
First, I identify which routine is most difficult for the child, such as getting dressed, bedtime, or bathing. Then I break the routine into small, simple steps and give instructions one at a time. I will use gentle reminders for them to complete each task.

I just tell them it is time to bathe, which usually leads to a meltdown because there is no visual warning.

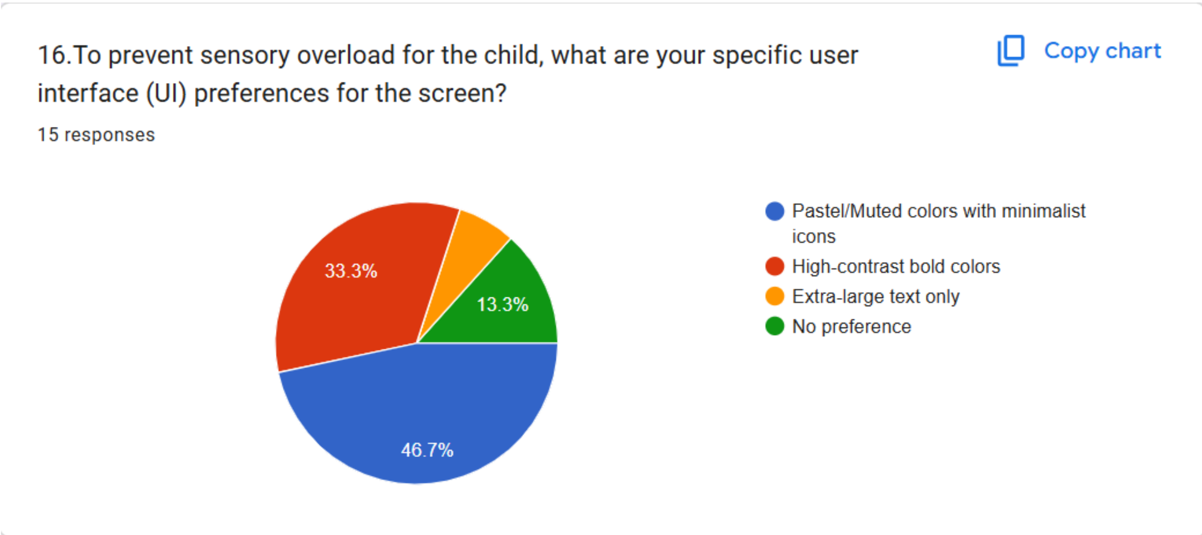
To understand the current AS-IS workflow and identify the main problems experienced by caregivers, respondents were asked to explain the methods they currently use to guide children through challenging daily routines. The responses show that caregivers still depend heavily on manual and less organized methods. Common methods include repeatedly giving verbal instructions, breaking routines into smaller and simpler tasks, and using visual tools such as whiteboards or printed schedules placed on walls. Although these methods help provide some structure, caregivers reported several difficulties. A lot of respondents mentioned that they must constantly remind, guide, and monitor the child throughout the routine to ensure tasks are completed properly. Some also shared that sudden changes or the absence of clear visual warnings can trigger emotional stress or emotional outbursts. These findings are important for the system analysis because they highlight the weaknesses of the current manual process. They also support the need for a “TO-BE” digital solution that can help organize routines automatically into simple visual steps, provide transition reminders, and reduce the workload placed on caregivers.



When respondents were asked to identify the most important feature for an Early Screening module, most emphasized the need for automatic result calculation. Out of the 15 respondents, 60% selected an instant automated risk calculation feature as their top priority. Meanwhile, 20% preferred a printable PDF report that could be easily shared with medical professionals, while another 20% highlighted the importance of having a built in list of nearby specialist clinics. These findings show that users place the greatest importance on fast and accurate assessment results. From a system design perspective, this indicates that the proposed “TO-BE” system should prioritize a reliable processing system capable of generating immediate screening results. In addition, supporting features such as PDF report generation and clinic location tracking should also be included to provide a more complete and user friendly experience.



To determine the main priorities for the daily routine module, the survey gathered 16 responses evaluating the usefulness of different interactive features. The feedback highlights a strong preference for transition support tools, with 68.8% of respondents marking visual countdown timers for task transitions as their main focus. Meanwhile, food texture and allergy logging became the second most requested feature, capturing exactly 25% of the responses. The remaining small percentage of users selected sensory friendly bedtime storytelling and audio features. Based on the survey results, these results indicate that the "TO-BE" system must place the main focus on real-time visual alerts and timing mechanics within the routine tracker to ease daily transitions, while secondary features should focus on dietary logging capabilities.



To address sensory and accessibility concerns, the survey gathered 15 responses about preferred user interface (UI) design styles. The results show that most respondents favor a softer and calmer visual design. About 46.7% preferred pastel or muted colors combined with simple icons, as these are seen as more comfortable and less stressful for children. However, clearer visuals was also identified as an important factor. Around 33.3% of respondents preferred high-contrast and brighter colors because they make information easier to notice and understand. The remaining responses were divided between users who preferred extra large text with minimal visual elements (6.7%) and those who had no specific preference (13.3%). These findings highlight that users have different sensory and visual needs. From a system design perspective, this creates an important additional system requirement for the proposed “TO-BE” system. Instead of using only one fixed design, the system should provide adjustable visual settings, such as a calming theme with soft colors and another brighter theme with stronger contrast,to better support different user preferences and sensory preferences.

<u>Interview</u>

To better understand how the current manual process works and what needs to be included in the system, qualitative methods were used in conjunction with surveys. An interview was conducted with TS. Nurasyikin, the subject matter expert, during the data collection phase. This session gave our team a clearer view of the current workflow, key challenges and what is expected from the proposed Autism Early Screening and Daily Routine Management System.

Question 1: What is the problem, solution and requirement of the system?

Answer: The main problem is that the current approach depends on a lot of manual and inconsistent tracking methods, which might be leading to missing or incorrect data. The suggested solution is to have a centralized digital platform that brings screening and daily routine management together in one system. Making the system simpler and easier to use is the most important requirement, so that parents and educators can enter and access the sensitive behavioral data quicker and safely.

Question 2: How do you handle the current process?

Answer: As of right now, the process is usually done using physical notes and different communication methods. Caregivers and educators usually write down the observations or daily routines in logbooks or on whiteboards. When a child moves between caregiver and teacher, the information is often shared verbally or through handing over notes, which can sometimes cause important details to be missed or misunderstood.

Question 3: Do you have a workflow?

Answer: There is a workflow, but it is not that consistent. It usually starts with a caregiver observing the child and writing down what they saw using whatever method is available at that time. However, the process is not well structured after that, especially during handovers, since there is no proper system or reminder to make sure the next person reviews the information before continuing the routine.

Question 4: Is there any standard format for reporting?

Answer: At the moment, there is no standard format for reporting. Different caregivers, schools, and medical staff use different styles and templates. Some write simple notes, while others use basic checklists. Because of this, it becomes difficult to combine all the information into a proper long-term report when referring the child for medical evaluation.

10.0 Requirement Analysis

<u>Problem Identification:</u>

- **No early ASD screening:** Parents who suspect their child have autism currently have no guided, structured tool to help them identify early signs. They rely entirely on personal observation or wait for a referral from a general practitioner, which is a slow and unstructured process.
- **No dedicated system for managing daily routines:** Parents of individuals with ASD currently track feeding,sleep,and hygiene routines manually through notebooks, physical whiteboard, general mobile calendar, and human memory. This method is inconsistent, and does not account for the sensory sensitivities and behavioural progression needs specific to individuals with ASD.
- **Fragmented support between parents and healthcare providers:** There is no single platform connecting parental screening concerns with daily routine data and clinical consultations. Healthcare providers receive only reports during appointments, leaving them without the structured data needed for practitioners to make great recommendations.

<u>User Needs:</u>

- **Parents/Caregivers:** A simple, guided tool to check early ASD behavioural signs at home without waiting for a report.
- **Parents/Caregivers:** A dedicated system to track and manage daily routine such as feeding, sleep and hygiene routines in a structured, sensory-friendly way.
- **Healthcare:** Access to structured, pre-consultation data from caregivers covering both screening responses and routine history.

<u>Current Challenges</u>

Information and Access Challenges:

- Parents have no starting point when they first suspect ASD because there is no tool that walks them through validated behavioural indicators.
- Information about ASD signs is scattered across unreliable online sources with no structured guidance.
- Reports from general practitioners to specialists can span weeks to months, delaying diagnosis.

Daily Management Challenges:

- Caregivers manually track feeding preferences, sleep patterns, and hygiene activities through informal methods.
- Scheduling apps do not accommodate sensory sensitivities, visual communication needs, or behavioural progression.
- Inconsistent tracking leads to incomplete data that is difficult to reference or share.

Clinical Consultation Challenges:

- Healthcare providers spend a significant portion of consultation time collecting basic background information.
- Providers have no access to structured routine data between appointments.

- Clinical recommendations are made based on incomplete or caregiver reports.
- Follow-up appointments are required just to collect updated routine information that could otherwise be tracked digitally.

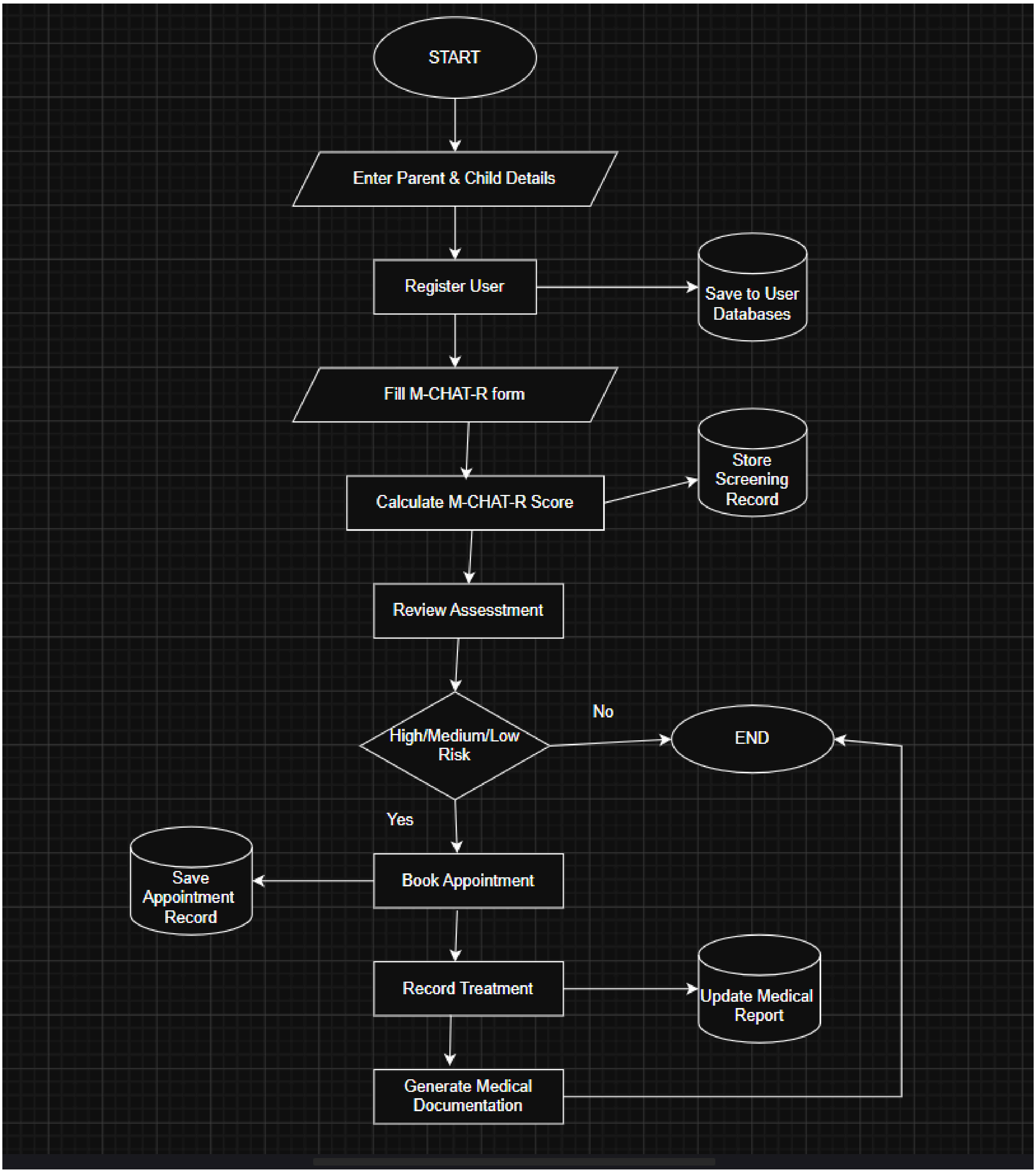
10.1 Current Business Process

Parent / Caregiver — Current Process (Without System)

1. Parents notice unusual behaviour or developmental differences in their child.
2. Parents search informally online or consult friends and family for information about ASD signs.
3. Parents book an appointment with a general practitioner to raise concerns.
4. GP refers parents to a developmental paediatrician or specialist so wait time can range from weeks to months.
5. While waiting, the parent manually tracks the child's daily routine through notebooks, phone notes, or memory.
6. Parents bring informal handwritten notes or verbal summaries to the specialist consultation.
7. Significant consultation time is spent by the specialist collecting basic background information from the parent verbally.
8. Specialists make recommendations based on incomplete or data.
9. Parents continue manual tracking between appointments with no structured system.

Healthcare Provider — Current Process (Without System)

1. The healthcare provider receives a report from the parent during consultation.
2. The healthcare provider has no access to structured screening data or routine history prior to the appointment.
3. The healthcare provider spends 10 to 20 minutes per consultation collecting background information that could have been pre-recorded.
4. The healthcare provider makes clinical recommendations based on fragmentary information..
5. Provider schedules follow-up appointments to collect updated routine information rather than accessing it digitally.


START
Enter Parent & Child Details
Register User
Save to User Databases
Fill M-CHAT-R form
Calculate M-CHAT-R Score
Store Screening Record
Review Assesstment
High/Medium/Low Risk
No
END
Yes
Book Appointment
Save Appointment Record
Record Treatment
Update Medical Report
Generate Medical Documentation

10.2 Functional Requirements

Context Diagram:

Process	Input	Output
ASD early screening system	Child details User details M-CHAT-R form request M-CHAT-R responses M-CHAT-R form Appointment details Request appointment Check appointment Medical details	Evaluation result Medical result

Diagram 0:

Process	Input	Output
User registration	Child details User details	User profile
M-CHAT-R assessment	M-CHAT-R form request M-CHAT-R responses M-CHAT-R form	Assessment result
Assessment result	Assessment data	Assessment summaries M-CHAT-R scoring Evaluation result
Book appointment	Appointment details Request appointment Check appointment Appointment schedule	Booking slot
Record treatment	Medical details	Medical update Medical result
Medical documentation	Medical report	Medical data

Child Diagram: User Registration

Process	Input	Output
Receive patient information	Child details User details	Patient information
Verify information	Patient information	Verified information
Create patient record	Verified information	Patient record
Store registration data	Patient record	User profile

Child Diagram: Screening assessment

Process	Input	Output
Provide screening form	M-CHAT-R request M-CHAT-R form	Blank M-CHAT-R form
Fill screening form	Blank M-CHAT-R form M-CHAT-R responses	M-CHAT-R responses
Collect screening record	M-CHAT-R responses	Verified M-CHAT-R form
Store screening data	Verified M-CHAT-R form	Assessment result

Child Diagram: Evaluation process

Process	Input	Output
Retrieve screening data	Assessment data	Screening responses data
Retrieve patient information	Screening responses data	Patient profile data
Analyze screening results	Patient profile data	Risk level
Generate evaluation record	Risk level	Evaluation report draft
Store result	Evaluation report draft	Assessment summaries Evaluation result

Child Diagram: Book appointment

Process	Input	Output
Schedule appointment	Request appointment	Appointment request data
Check patient record	Appointment request data Appointment details	Verified patient info
Check availability schedule	Verified patient info	Available time slot
Confirm appointment	Available time slot	Confirmed appointment details
Store appointment	Confirmed appointment details Check appointment Appointment schedule	Booking slot

Child Diagram: Record treatment

Process	Input	Output
Assess patient condition	Medical details	Diagnosis recommendation
Record treatment	Diagnosis recommendation	Treatment details
Update treatment	Treatment details	Updated treatment record
Store treatment	Updated treatment record	Medical update Medical result

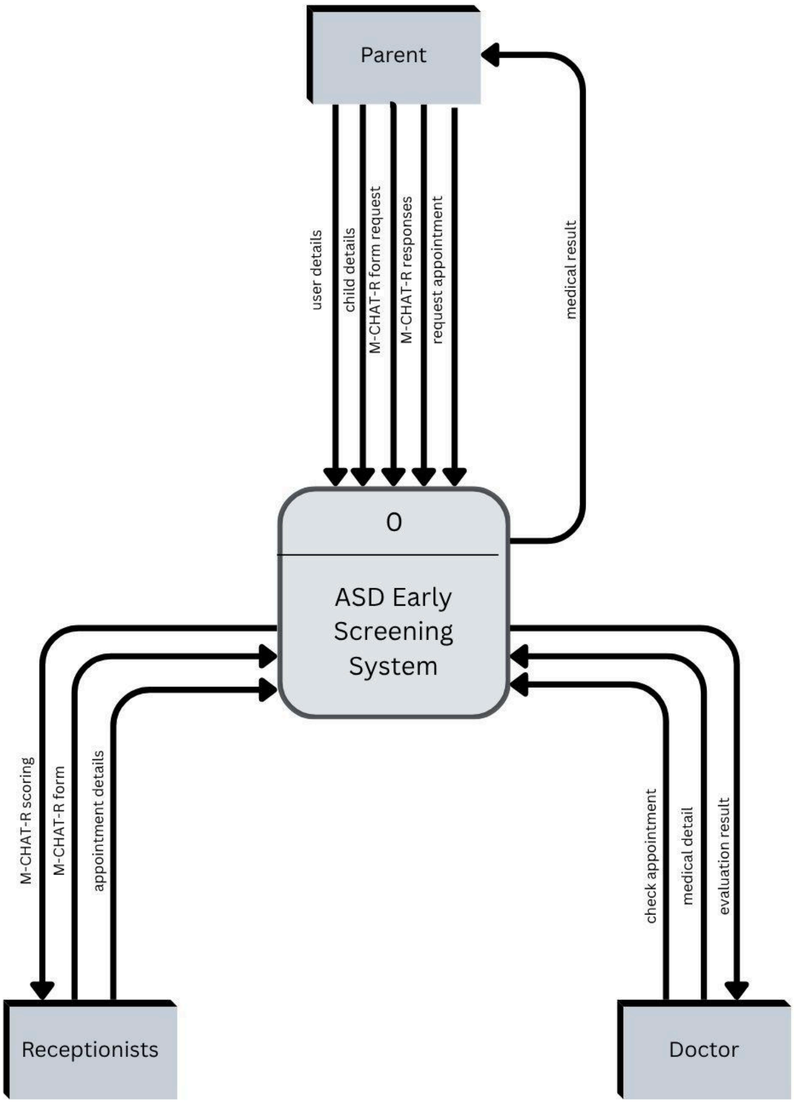
10.3 Non-functional Requirements

Based on the analysis from the questionnaire feedback, several non-functional requirements were identified for the proposed system. These requirements are important to ensure the system is efficient for user experience.

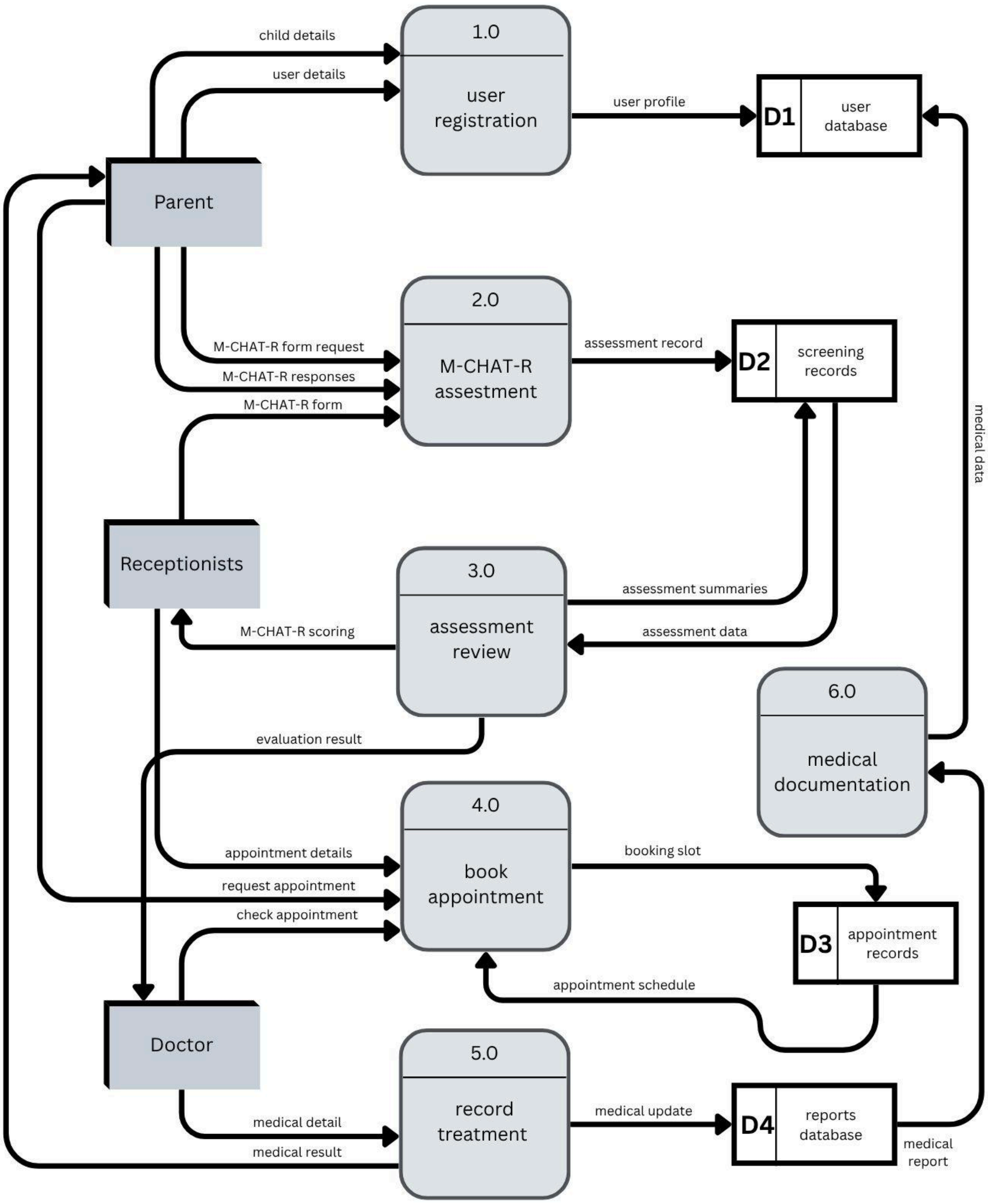
No	Category	Requirement	Control
1	Performance	Fast and responsive system for autism screening and result generation.	Regularly monitor the system performance and improve the loading speed to ensure the system runs smoothly.
2	Security	Protect users personal information from unauthorized access.	Secure login authentication, data encryption and restricted access to screening records.
3	Usability	Provide a simple and user-friendly interface for users.	Conduct usability testing and collect user feedback for system improvement.
4	Accessibility	Ensure the system is easy for all users to understand and use.	Implement simple navigation, readable fonts and AI support features to assist users using the mobile app or web platform.
5	Reliability	Ensure the system operates consistently with minimal downtime and errors.	Perform regular system maintenance and fix errors to ensure the system runs smoothly.
6	Regulatory Compliance	Ensure compliance with data privacy and protection regulations.	Apply privacy policies and secure data storage practices.

10.4 Logical Data Flow Diagram (DFD)

10.4.1 AS-IS Context Diagram

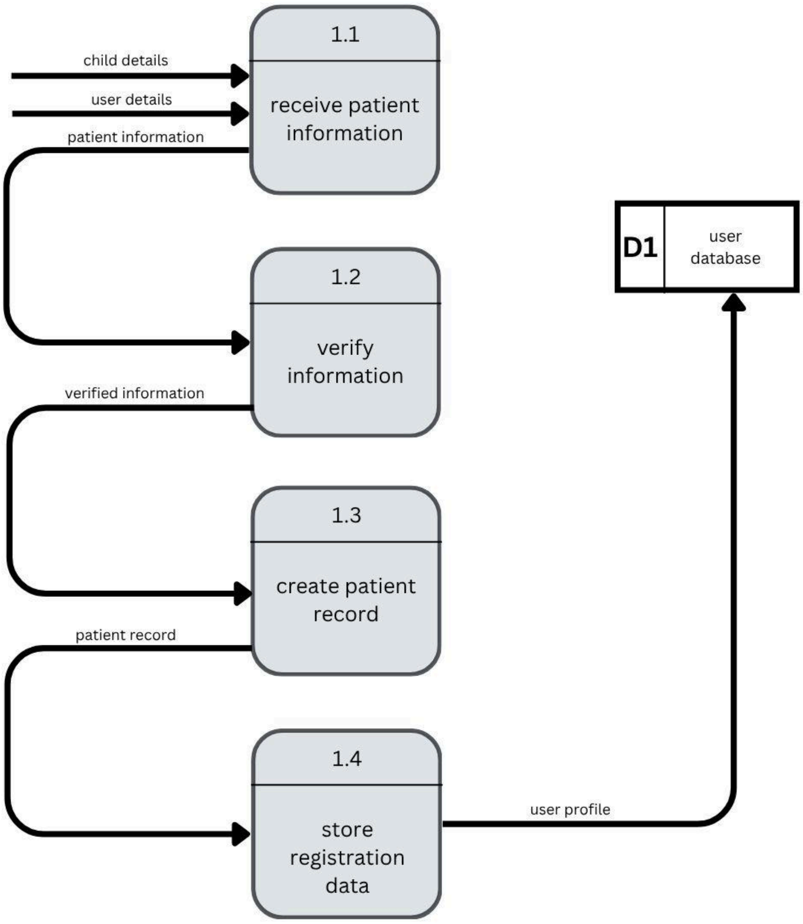


10.4.2 AS-IS Diagram 0

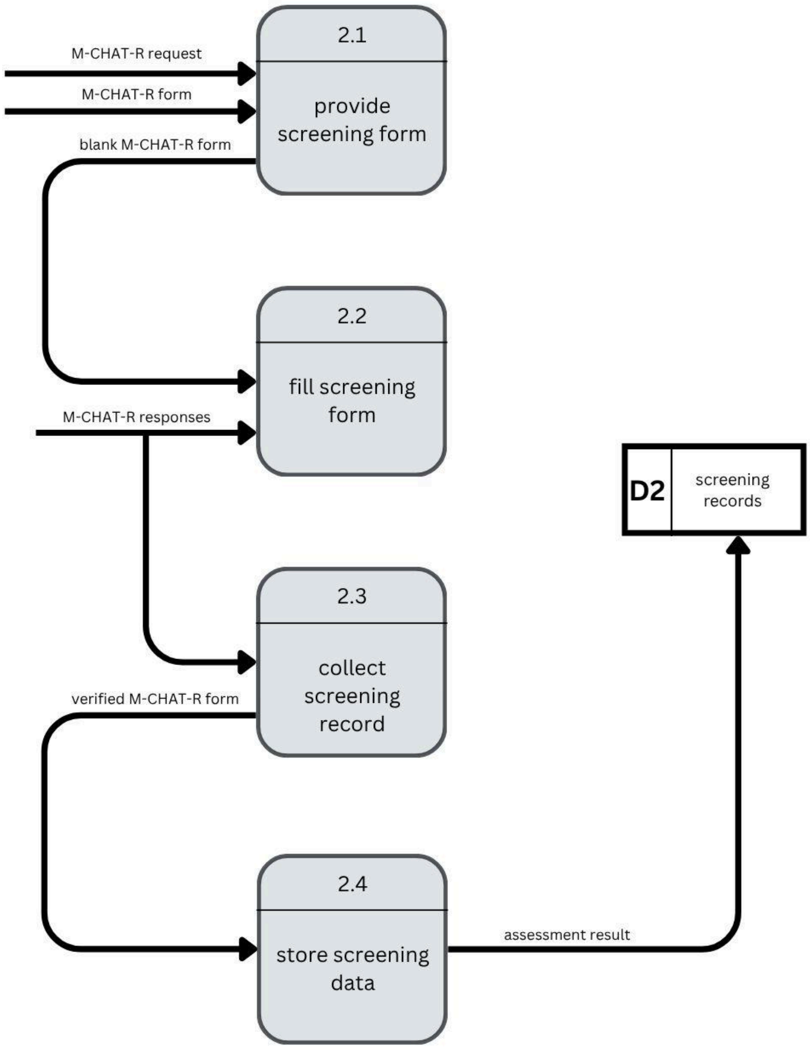


10.4.2 AS-IS Child Diagram

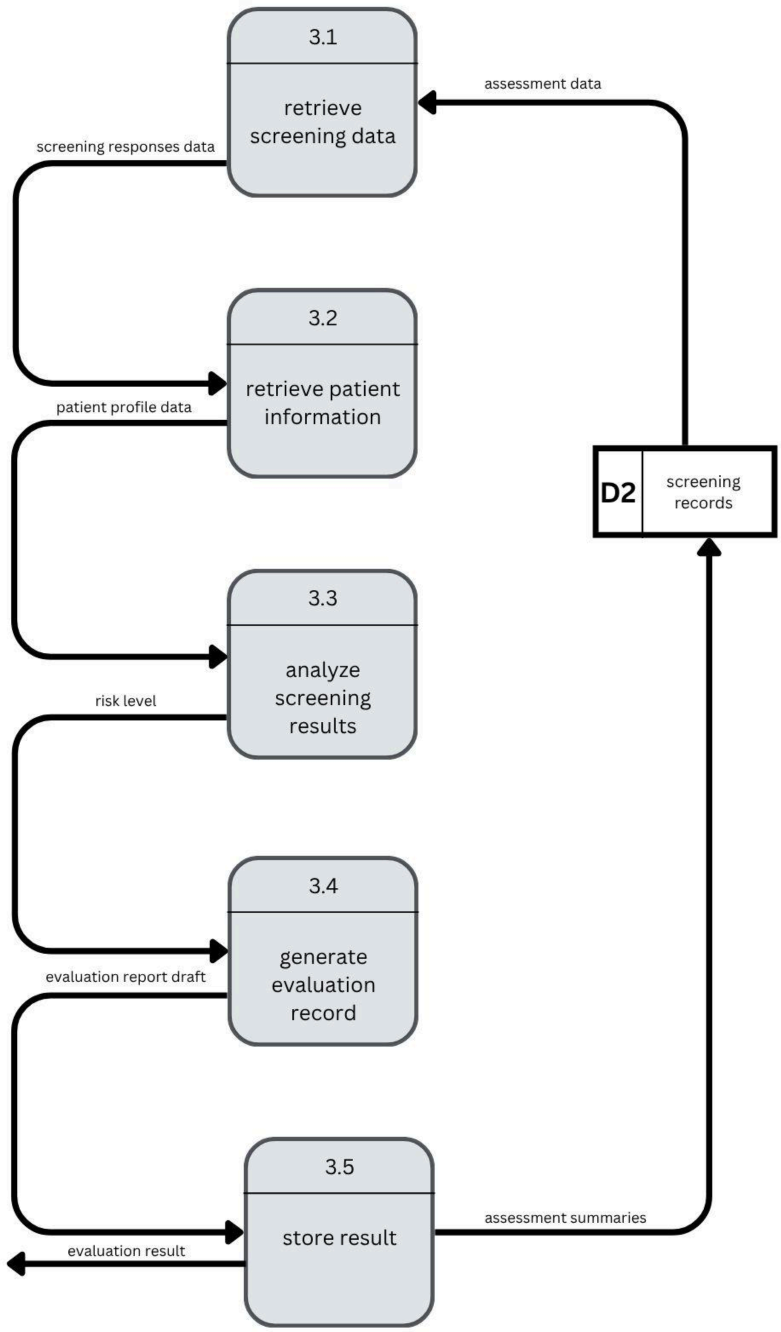
- User Registration



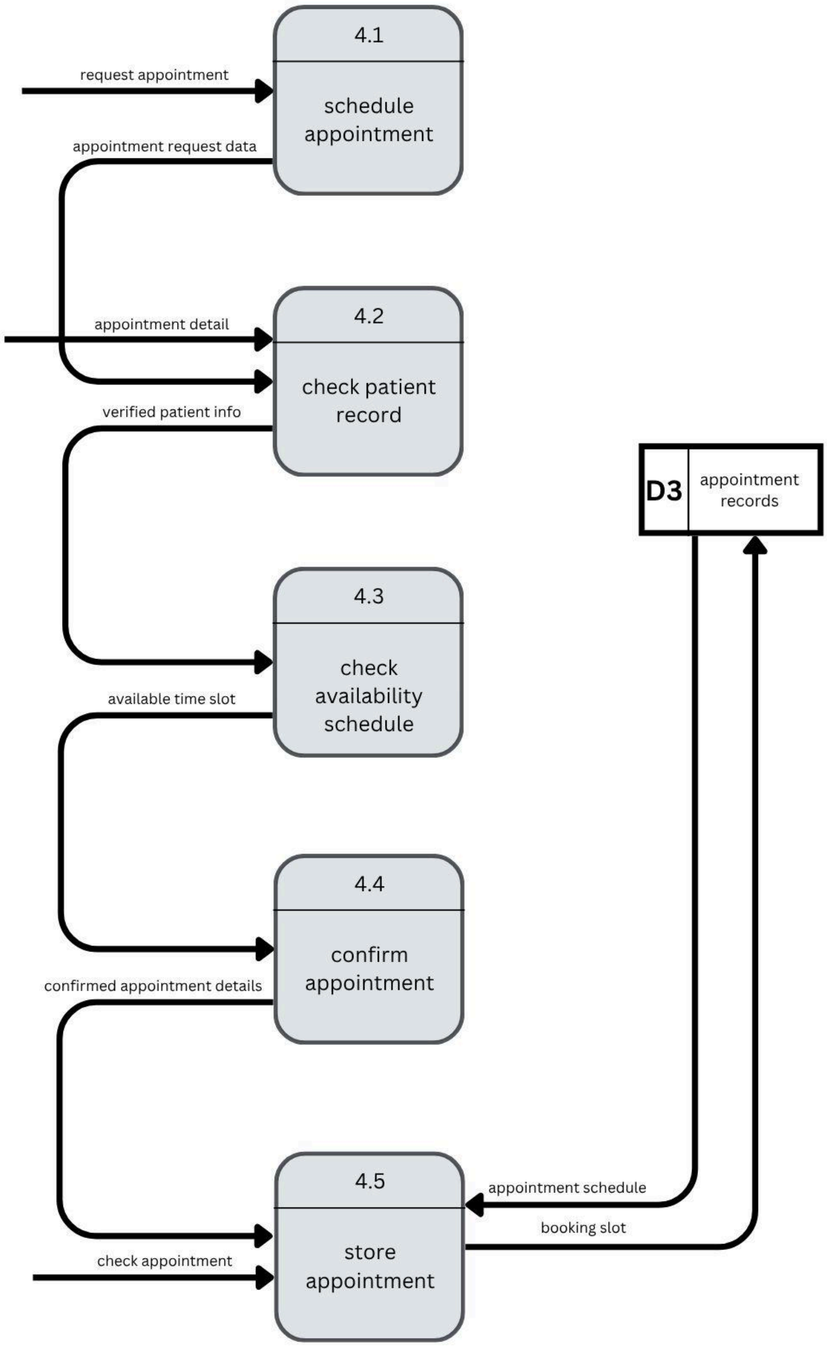
- M-CHAT-R Assessment



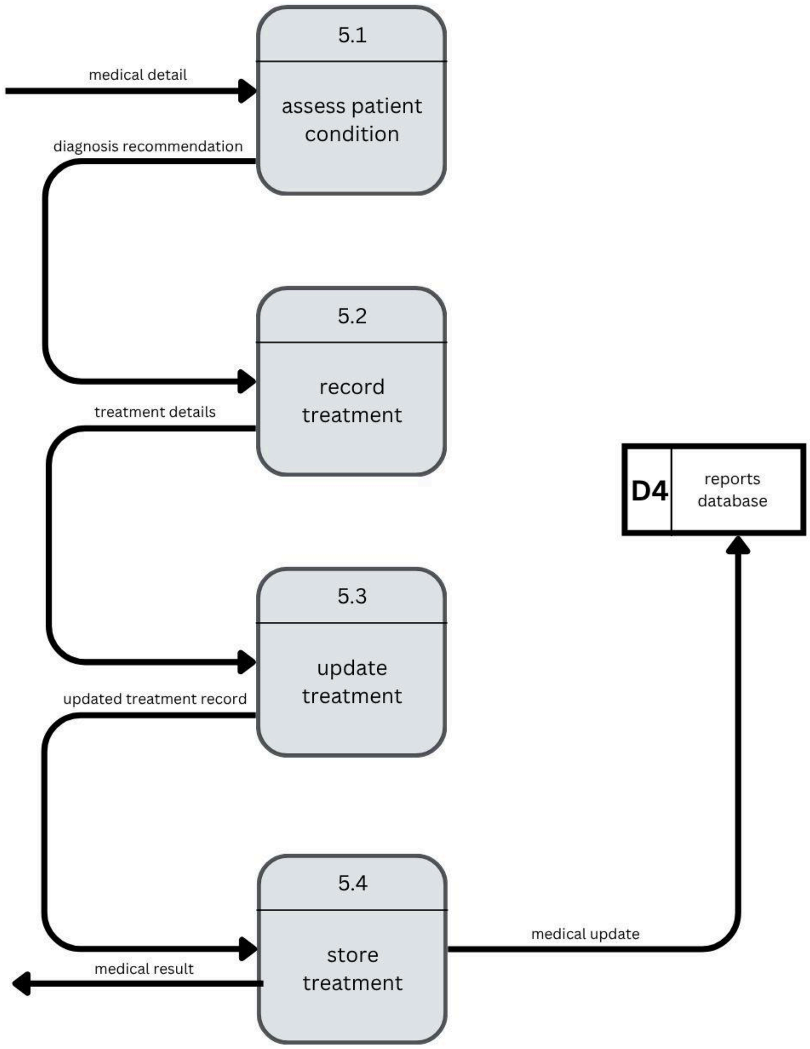
- Assessment Review



- Book Appointment



- Record Treatment



11.0 Summary of Requirement Analysis

The requirement analysis phase was conducted to identify the current problems, user needs and system requirements for the proposed Autism Early Screening Support System and Daily Routine Management System. Based on the information gathered through interviews and questionnaires of the current workflow, parents and caregivers currently rely on manual and unstructured methods such as notebooks, whiteboards, phone notes and memory to manage autism screening information and daily routines. This causes inconsistent tracking, missing records, delayed diagnosis and difficulty in sharing information with healthcare providers. The analysis also showed that there is no centralized platform that integrates autism early screening and daily routine management in one system. Parents often depend on unreliable online information or lengthy referral processes before receiving professional assessment. Parents also struggle to organize and track important information consistently which makes it difficult to provide accurate information for further consultation with healthcare providers.

From the questionnaires and interview findings, users expressed the need for a simple, structured and autism-friendly digital platform. There are several functional requirements identified include user registration, M-CHAT-R screening assessment, automatic scoring and result generation and daily routine tracking. Users also emphasised the importance of visual countdown timers, reminder system, PDF report generation and customizable sensory-friendly interfaces. In addition, non-functional requirements were identified to ensure system quality and usability. These include fast system performance, secure data handling, accessibility features, reliability and compliance with data privacy regulations.

Overall, the requirement analysis confirms that the proposed system is necessary to improve autism early screening awareness, help in daily routine management and provide a more organized and centralized platform that ease both parents and healthcare providers. The findings from this phase give as reference for the system design phase of the proposed TO-BE system.

12.0 References

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[5] Russell, A. S., McFayden, T. C., McAllister, M., Liles, K., Bittner, S., Strang, J. F., & Harrop, C. (2024). Who, when, where, and why: A systematic review of "late diagnosis" in autism. *Autism Research*, *18*(1). https://doi.org/10.1002/aur.3278

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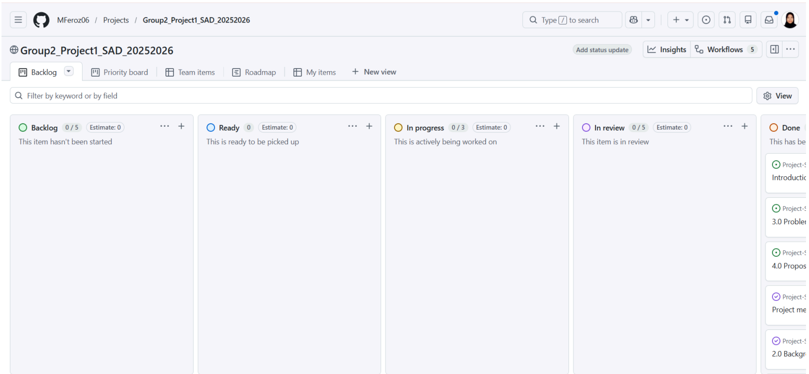
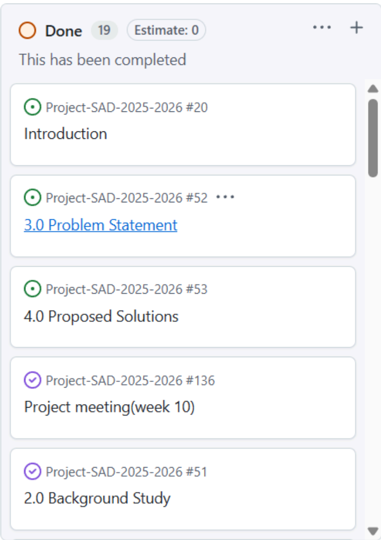
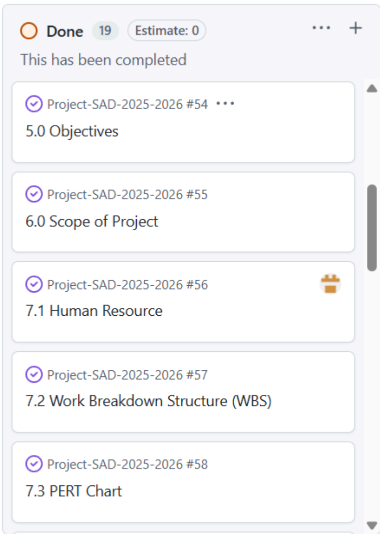
https://marf.org.my/establishing-routines-in-children-with-autism/

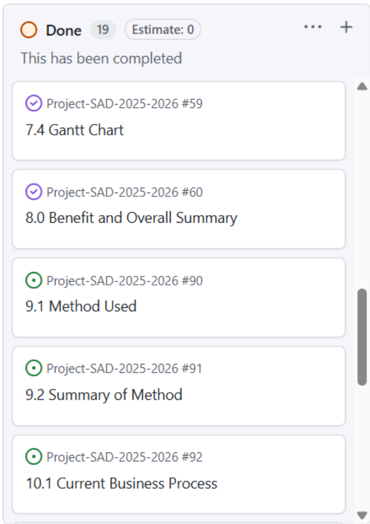
13.0 Appendix

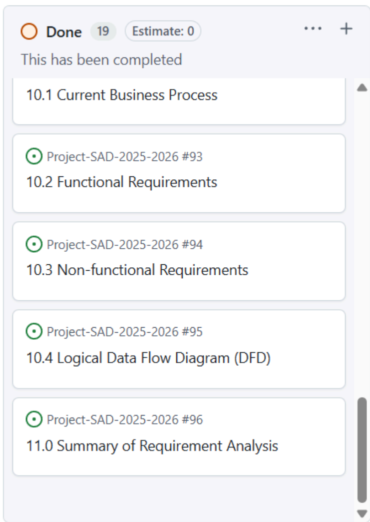
1. Github





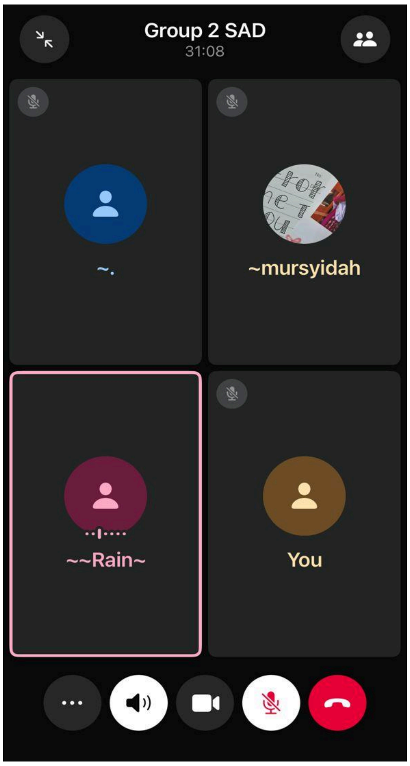






Link: https://github.com/users/MFeroz06/projects/48

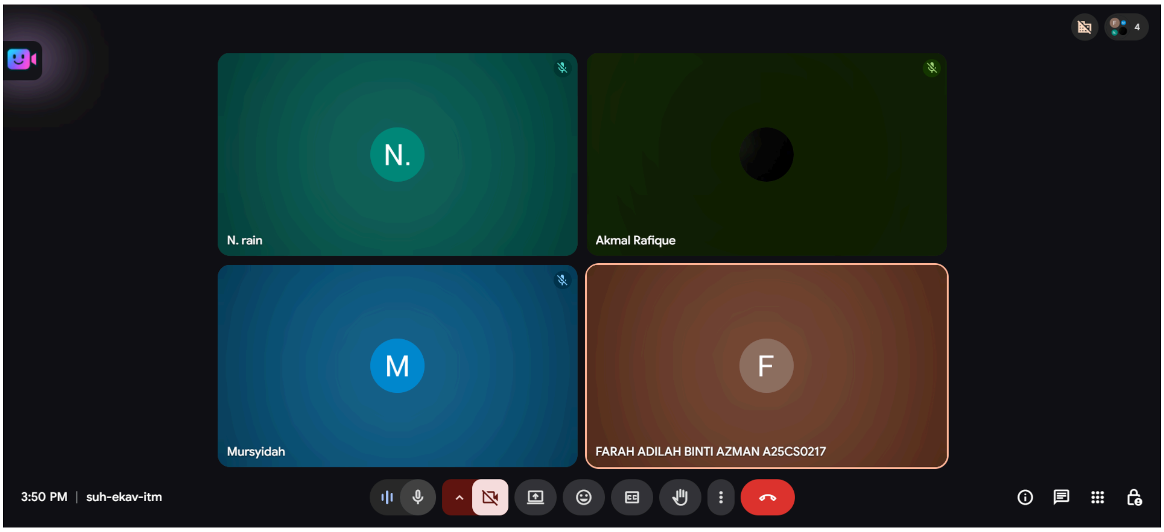
2. Meeting 1: 4/5/2026 (Monday)



Progress:

- Task distribution for phase 1 to all of the group members
- Decide a topic for this project
- Find resources on autism
- Plan next meeting on 14/5/2026 (Thursday)

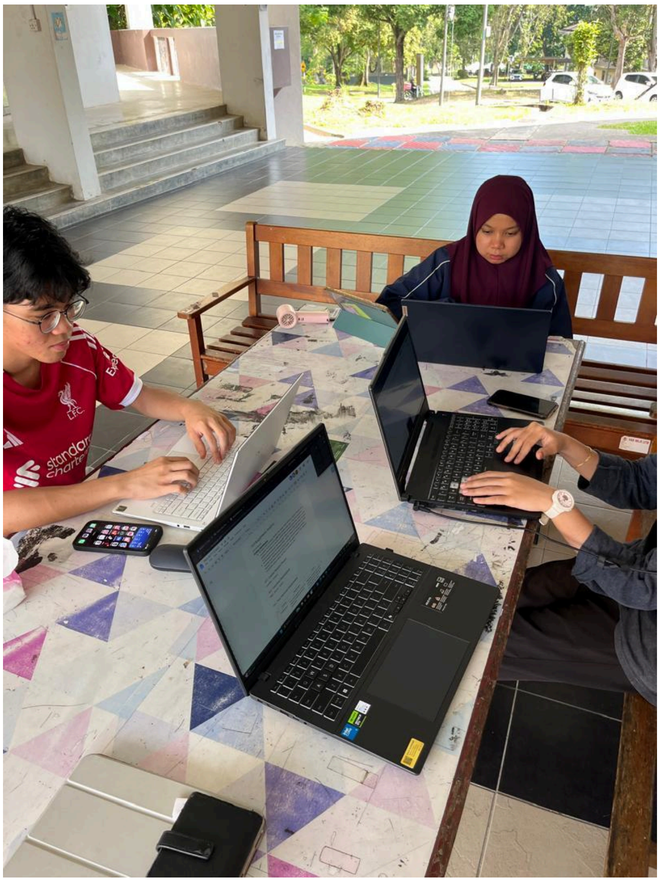
3. Meeting 2: 14/5/2026 (Thursday)



Progress:

- Update progress on phase 1
- Final review on phase 1
- Task distribution for phase 2
- Plan next meeting on 17/5/2026 (Sunday)

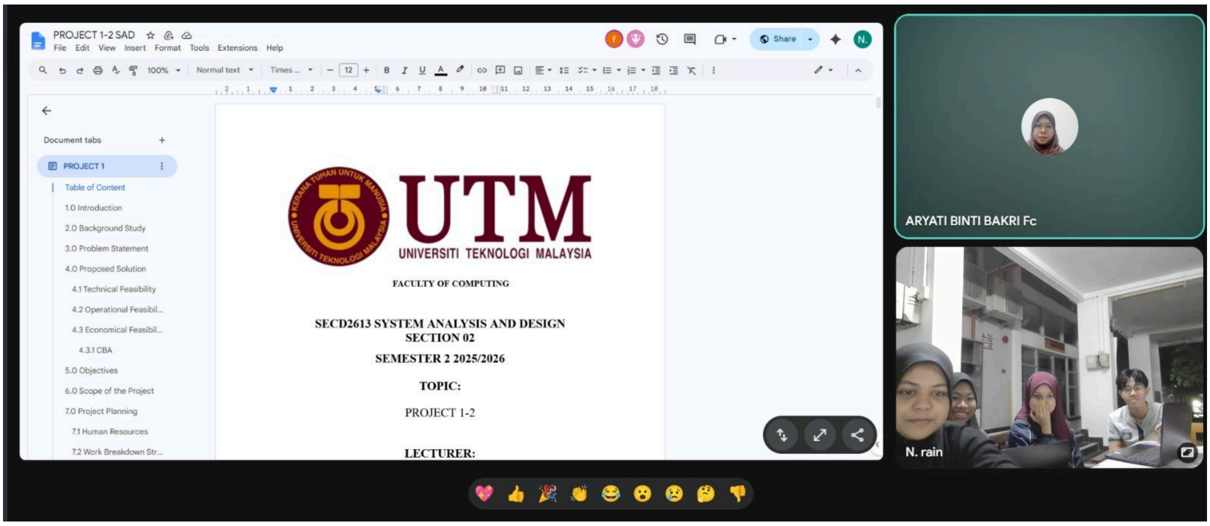
4. Meeting 3: 17/5/2026 (Sunday)



Progress:

- Update progress on phase 2
- Final review on phase 2
- Plan next meeting with Dr. Aryati on 19/5/2026 (Tuesday)

5. Meeting 4: 19/5/2026 (Tuesday)



Progress:

- Final review with Dr Aryati
- Make improvement for phase 1 on introduction, background study, feasibility, objective and WBS.
- Make improvement for phase 2 on citation, make a flowchart and include interview in the method used

6. Questionnaire: https://forms.gle/1ybRPY6RW1PtyWZ58